Problem 1

Solution:

(a) $P = \{-1, 0, 1, 2, 3\}$ is a partition of $[-1, 3]$ where

$$[x] = \begin{cases} -1 & \text{if } -1 \leq x < 0 \\ 0 & \text{if } 0 \leq x < 1 \\ 1 & \text{if } 1 \leq x < 2 \\ 2 & \text{if } 2 \leq x < 3 \\ 3 & \text{if } x = 3 \end{cases}$$

Therefore,

$$\int_{-1}^{3} [x]\, dx = 1(-1) + 1(0) + 1(1) + 1(2) = 2$$

(b) $P = \{-1, -\frac{1}{2}, \frac{1}{2}, \frac{3}{2}, \frac{5}{2}, 3\}$ is a partition of $[-1, 3]$ where

$$\left[x + \frac{1}{2}\right] = \begin{cases} -1 & \text{if } -1 \leq x < -\frac{1}{2} \\ 0 & \text{if } -\frac{1}{2} \leq x < \frac{1}{2} \\ 1 & \text{if } \frac{1}{2} \leq x < \frac{3}{2} \\ 2 & \text{if } \frac{3}{2} \leq x < \frac{5}{2} \\ 3 & \text{if } \frac{5}{2} \leq x \leq 3 \end{cases}$$

Therefore,

$$\int_{-1}^{3} \left[x + \frac{1}{2}\right]\, dx = \frac{1}{2}(-1) + 1(0) + 1(1) + 1(2) + \frac{1}{2}(3) = 4$$

(c) $P = \{-1, -\frac{1}{2}, 0, \frac{1}{2}, 1, \frac{3}{2}, 2, \frac{5}{2}, 3\}$ is a partition of $[-1, 3]$ where

$$[x] + \left[x + \frac{1}{2}\right] = \begin{cases} -2 & \text{if } -1 \leq x < -\frac{1}{2} \\ -1 & \text{if } -\frac{1}{2} \leq x < 0 \\ 0 & \text{if } 0 \leq x < \frac{1}{2} \\ 1 & \text{if } \frac{1}{2} \leq x < 1 \\ 2 & \text{if } 1 \leq x < \frac{3}{2} \\ 3 & \text{if } \frac{3}{2} \leq x < 2 \\ 4 & \text{if } 2 \leq x < \frac{5}{2} \\ 5 & \text{if } \frac{5}{2} \leq x < 3 \\ 6 & \text{if } x = 3 \end{cases}$$

Therefore,

$$\int_{-1}^{3} [x] + \left[x + \frac{1}{2}\right] \, dx = \frac{1}{2}\left[-2 + -1 + 0 + 1 + 2 + 3 + 4 + 5\right] = 6$$

(d) By the homogeneous property and part (a)

$$\int_{-1}^{3} 2[x] \, dx = 2\int_{-1}^{3} [x] \, dx = 2(2) = 4$$

(e) By the expansion/contraction theorem and interval additivity theorem

$$\int_{-1}^{3} [2x] \, dx = \frac{1}{2}\int_{-2}^{6} [x] \, dx = \frac{1}{2}\left[\int_{-2}^{-1} [x] \, dx + \int_{-1}^{3} [x] \, dx + \int_{3}^{6} [x] \, dx\right]$$

$P = \{-2, -1\}$ is a partition of $[-2, -1]$ where

$$[x] = \Big\{ -2 \quad \text{if } -2 \leq x < -1$$

$Q = \{3, 4, 5, 6\}$ is a partition of $[3, 6]$ where

$$[x] = \begin{cases} 3 & \text{if } 3 \leq x < 4 \\ 4 & \text{if } 4 \leq x < 5 \\ 5 & \text{if } 5 \leq x < 6 \end{cases}$$

Therefore,

$$\frac{1}{2}\left[\int_{-2}^{-1} [x] \, dx + \int_{-1}^{3} [x] \, dx + \int_{3}^{6} [x] \, dx\right] = \frac{1}{2}\left[[1(-2)] + [2] + [1(3) + 1(4) + 1(5)]\right] = 6$$

(f) By the reflection property

$$\int_{-1}^{3} [-x] \, dx = \int_{-3}^{1} [x] \, dx$$

$P = \{-3, -2, -1, 0, 1\}$ is a partition of $[-3, 1]$ where

$$[x] = \begin{cases} -3 & \text{if } -3 \leq x < -2 \\ -2 & \text{if } -2 \leq x < -1 \\ -1 & \text{if } -1 \leq x < 0 \\ 0 & \text{if } 0 \leq x < 1 \\ 1 & \text{if } x = 1 \end{cases}$$

Therefore,

$$\int_{-3}^{1} [x]\ dx = 1(-3) + 1(-2) + 1(-1) + 1(0) = -6$$

□

Problem 2

Solution:

Let

$$s(x) = \begin{cases} \frac{5}{2} & \text{if } 0 \leq x \leq 2 \\ -1 & \text{if } 2 < x \leq 5 \end{cases}$$

Then

$$\int_0^2 s(x)dx = 2\left(\frac{5}{2}\right) = 5$$

$$\int_2^5 s(x)dx = 3(-1) = -3$$

and by the interval additivity theorem

$$\int_0^5 s(x)dx = \int_0^2 s(x)dx + \int_2^5 s(x)dx = 5 - 3 = 2$$

□

Problem 3

Solution:

By the reflection property, we have

$$\int_a^b [x]\ dx = \int_{-b}^{-a} [-x]\ dx$$

$$\int_a^b [-x]\ dx = \int_{-b}^{-a} [x]\ dx$$

Therefore,

$$\int_a^b [x]\ dx + \int_a^b [-x]\ dx = \int_{-b}^{-a} [-x]\ dx + \int_{-b}^{-a} [x]\ dx$$

By the additive theorem of integrals

$$\int_{-b}^{-a} [-x]\ dx + \int_{-b}^{-a} [x]\ dx = \int_{-b}^{-a} [-x] + [x]\ dx$$

Here, recall that

$$[-x] = \begin{cases} -[x] & \text{if } x \text{ is an integer} \\ -[x] - 1 & \text{otherwise} \end{cases}$$

Hence,

$$s(x) = [-x] + [x] = \begin{cases} 0 & \text{if } x \text{ is an integer} \\ -[x] - 1 + [x] = -1 & \text{otherwise} \end{cases}$$

Let $P = -b = k_0 < k_1 < \ldots < k_n = -a$ be a partition of $[-b, -a]$ where the interior k_i (i.e. $k_1, k_2, \ldots, k_{n-1}$) are all integers. Then, we have

$$\int_{-b}^{-a} [-x] + [x]\ dx = \sum_{i=1}^{n} (-1)(k_i - k_{i-1}) = -\sum_{i=1}^{n} [k_i - k_{i-1}] = -(k_n - k_0) = -(-a + b) = a - b$$

□

Problem 4

Solution:

(a) If n is a positive integer then $P = \{0, 1, 2, \ldots, n-1, n\}$ is a partition of $[0, n]$ where

$$[t] = \begin{cases} 0 & \text{if } 0 \leq t < 1 \\ 1 & \text{if } 1 \leq t < 2 \\ \vdots & \\ n-1 & \text{if } (n-1) \leq t < n \end{cases}$$

Hence,

$$\int_0^n [t]\ dt = 1(0) + 1(1) + 1(2) + \cdots 1(n-1) = \frac{n(n-1)}{2}$$

(b) Omitted. Use https://www.desmos.com/ instead.

□

Problem 5

Solution:

(a) $P = \{0, 1, \sqrt{2}, \sqrt{3}, 2\}$ is a partition of $[0, 2]$ where

$$\left[t^2\right] = \begin{cases} 0 & \text{if } 0 \leq t < 1 \\ 1 & \text{if } 1 \leq t < \sqrt{2} \\ 2 & \text{if } \sqrt{2} \leq t < \sqrt{3} \\ 3 & \text{if } \sqrt{3} \leq t < 2 \end{cases}$$

Hence,

$$\int_0^2 \left[t^2\right] \, dt = 1(0) + (\sqrt{2} - 1)(1) + (\sqrt{3} - \sqrt{2})(2) + (2 - \sqrt{3})(3) = 5 - \sqrt{2} - \sqrt{3}$$

(b) By the additivity of intervals theorem

$$\int_{-3}^3 \left[t^2\right] \, dt = \int_{-3}^0 \left[t^2\right] \, dt + \int_0^3 \left[t^2\right] \, dt$$

Now, by the reflection property, we have

$$\int_{-3}^0 \left[t^2\right] \, dt + \int_0^3 \left[t^2\right] \, dt = \int_0^3 \left[(-t)^2\right] \, dt + \int_0^3 \left[t^2\right] \, dt = 2\int_0^3 \left[t^2\right] \, dt$$

Here, we can apply the additivity of intervals theorem again to obtain

$$2\int_0^3 \left[t^2\right] \, dt = 2\left[\int_0^2 \left[t^2\right] \, dt + \int_2^3 \left[t^2\right] \, dt\right]$$

From part (a), we know the first integral evaluates to $5 - \sqrt{2} - \sqrt{3}$. For the second, observe that

$$\left[t^2\right] = \begin{cases} 4 & \text{if } 2 \leq t < \sqrt{5} \\ 5 & \text{if } \sqrt{5} \leq t < \sqrt{6} \\ 6 & \text{if } \sqrt{6} \leq t < \sqrt{7} \\ 7 & \text{if } \sqrt{7} \leq t < \sqrt{8} \\ 8 & \text{if } \sqrt{8} \leq t < 3 \end{cases}$$

so we have

$$
\begin{aligned}
&2\left[\int_0^2 \left[t^2\right]\ dt + \int_2^3 \left[t^2\right]\ dt\right] \\
&= 2\left[(5-\sqrt{2}-\sqrt{3}) + \left(\int_2^3 \left[t^2\right]\ dt\right)\right] \\
&= (10-2\sqrt{2}-2\sqrt{3}) + 2[4(\sqrt{5}-2) + 5(\sqrt{6}-\sqrt{5}) + 6(\sqrt{7}-\sqrt{6}) + 7(\sqrt{8}-\sqrt{7}) + 8(3-\sqrt{8})] \\
&= 42 - 6\sqrt{2} - 2\sqrt{3} - 2\sqrt{5} - 2\sqrt{6} - 2\sqrt{7}
\end{aligned}
$$

□

Problem 6

Solution:

(a) Omitted since it's almost identical to 4(a).

(b) Omitted. Use https://www.desmos.com/ instead.

(c) Since $[x] \leq x < [x]+1$ we have $0 \leq x - [x] < 1$. Also, $x = [x] + (x - [x])$.

Then, observe that $P = \{0, 1, 2, \ldots, [x], [x] + (x - [x])\}$ is a partition of $[0, x]$.

And by the interval additivity theorem, we have

$$\int_0^x [t]^2\ dt = \int_0^{[x]} [t]^2\ dt + \int_{[x]}^{[x]+(x-[x])} [t]^2\ dt$$

Putting it all together, we see

$$\int_0^x [t]^2\ dt = \left[\frac{[x]([x]-1)(2[x]-1)}{6}\right] + (x-[x])[x]^2$$

Now, we want the following, where the left side is just the above cleaned up

$$x[x]^2 - \frac{2[x]^3}{3} - \frac{[x]^2}{2} + \frac{[x]}{6} = 2x - 2$$

This can be written as

$$g(x) = x([x]^2 - 2) - \frac{2[x]^3}{3} - \frac{[x]^2}{2} + \frac{[x]}{6} + 2$$

Finding positive roots via https://www.desmos.com gives $x = 1, \frac{5}{2}$

□

Problem 7

Solution:

(a) $P = \{0, 1, 4, 9\}$ is a partition of $[0, 9]$ where

$$[\sqrt{t}] = \begin{cases} 0 & \text{if } 0 \leq t < 1 \\ 1 & \text{if } 1 \leq t < 4 \\ 2 & \text{if } 4 \leq t < 9 \end{cases}$$

so we have

$$\int_0^9 [\sqrt{t}]\, dt = 1(0) + 3(1) + 5(2) = 13$$

(b) If n is a positive integer, $P = \{0, 1^2, 2^2, \ldots, (n-1)^2, n^2\}$ is a partition of $[0, 9]$.

$$[\sqrt{t}] = \begin{cases} 0 & \text{if } 0 \leq t < 1 \\ 1 & \text{if } 1 \leq t < 4 \\ \vdots & \\ (n-1) & \text{if } (n-1)^2 \leq t < n^2 \end{cases}$$

so we have

$$\int_0^{n^2} [\sqrt{t}]\, dt = (1^2 - 0^2)(0) + (2^2 - 1^2)(1) + \cdots + (n^2 - (n-1)^2)(n-1) = \sum_{k=1}^{n} (k-1)(k^2 - (k-1)^2)$$

which simplifies to

$$\begin{aligned} \sum_{k=1}^{n} (k-1)(k^2 - (k-1)^2) &= \sum_{k=1}^{n} (k-1)(2k-1) \\ &= \sum_{k=1}^{n} 2k^2 - 3k + 1 \\ &= 2\left[\frac{n^3}{3} + \frac{n^2}{2} + \frac{n}{6}\right] - 3\left[\frac{n^2}{2} + \frac{n}{2}\right] + n \\ &= \frac{2n^3}{3} - \frac{n^2}{2} - \frac{n}{6} \\ &= \frac{n(n-1)(4n+1)}{6} \end{aligned}$$

□

Problem 8

Solution:

We need to show that for any step function f defined on the interval $[a, b+c]$

$$\int_a^b f(x)\ dx = \int_{a+c}^{b+c} f(x-c)\ dx \text{ (Theorem 1.7)} \iff \int_a^b f(x+c)\ dx = \int_{a+c}^{b+c} f(x)\ dx$$

We'll start with the forward direction. Let $g(x) = f(x+c)$. Then,

$$\int_a^b f(x+c)\ dx = \int_a^b g(x)\ dx = \int_{a+c}^{b+c} g(x-c)\ dx = \int_{a+c}^{b+c} f(x)\ dx$$

Now, the reverse case. Let $g(x) = f(x-c)$. Then,

$$\int_a^b f(x)\ dx = \int_a^b g(x+c)\ dx = \int_{a+c}^{b+c} g(x)\ dx = \int_{a+c}^{b+c} f(x-c)\ dx$$

□

Problem 9

Solution:

We need to show that for any step function f defined on the interval $[ka, kb]$

$$\int_{ka}^{kb} f\left(\frac{x}{k}\right)\ dx = k\int_a^b f(x)\ dx \text{ (Theorem 1.8)} \iff \int_{ka}^{kb} f(x)\ dx = k\int_a^b f(kx)\ dx$$

We'll start with the forward direction. Let $g(x) = f(kx)$. Then,

$$\int_{ka}^{kb} f(x)\ dx = \int_{ka}^{kb} g\left(\frac{x}{k}\right)\ dx = k\int_a^b g(x)\ dx = k\int_a^b f(kx)\ dx$$

Now, the reverse case. Let $g(x) = f\left(\frac{x}{k}\right)$. Then,

$$\int_{ka}^{kb} f\left(\frac{x}{k}\right)\ dx = \int_{ka}^{kb} g(x)\ dx = k\int_a^b g(kx)\ dx = k\int_a^b f(x)\ dx$$

□

Problem 10

Solution:

$$\begin{aligned} f(p) &= \int_0^p s(x)\,dx \\ &= 1(-1)^0(0) + 1(-1)^1(1) + 1(-1)^2(2) + \cdots + 1(-1)^{p-1}(p-1) \\ &= -1 + 2 - \cdots + (-1)^{p-1}(p-1) \end{aligned}$$

(a)

$$\begin{aligned} f(3) &= -1 + 2 && = 1 \\ f(4) &= -1 + 2 - 3 && = -2 \\ f(f(3)) &= 0 && = 0 \end{aligned}$$

(b)

Observe that $f(p+1) = f(p) + (-1)^p(p)$. Hence,

$$\begin{aligned} f(1) &= && = 0 \\ f(2) &= f(1) + (-1)^{2-1}(2-1) && = -1 \\ f(3) &= f(2) + (-1)^{3-1}(3-1) && = 1 \\ f(4) &= f(3) + (-1)^{4-1}(4-1) && = -2 \\ f(5) &= f(4) + (-1)^{5-1}(5-1) && = 2 \\ f(6) &= f(5) + (-1)^{6-1}(6-1) && = -3 \\ f(7) &= f(6) + (-1)^{7-1}(7-1) && = 3 \\ &\vdots \end{aligned}$$

Hence, we conjecture that $|f(2k)| = |f(2k+1)| = k$. It can be proved by induction (but the proof is omitted here). Thus, both $p = 14, 15$ will produce $|f(p)| = 7$.

□
□

Problem 11

Solution:

(a) Let $P = \{a = x_0, x_1, \ldots, x_j = b, x_{j+1}, \ldots, x_{n-1}, x_n = c\}$ be a partition of $[a, c]$.

$$\begin{aligned}\int_a^b s(x)\ dx + \int_b^c s(x)\ dx &= \sum_{k=1}^{j} s_k^3(x_k - x_{k-1}) + \sum_{k=j+1}^{n} s_k^3(x_k - x_{k-1}) \\ &= \sum_{k=1}^{n} s_k^3(x_k - x_{k-1}) \\ &= \int_a^c s(x)\ dx\end{aligned}$$

(b) Let $P = \{a = x_0, x_1, \ldots, x_n = b\}$ be a partition of $[a, b]$.

$$\begin{aligned}\int_a^b s(x) + t(x)\ dx &= \sum_{k=1}^{n} (s_k + t_k)^3(x_k - x_{k-1}) \\ &\neq \sum_{k=1}^{n} (s_k^3 + t_k^3)(x_k - x_{k-1}) \\ &= \int_a^b s(x)\ dx + \int_a^b t(x)\ dx\end{aligned}$$

since $(e + f)^3 = e^3 + f^3$ is not universally true for all reals e, f.

(c) Let $P = \{a = x_0, x_1, \ldots, x_n = b\}$ be a partition of $[a, b]$.

$$\begin{aligned}\int_a^b c \cdot s(x)dx &= \sum_{k=1}^{n} (cs_k)^3(x_k - x_{k-1}) \\ &= c^3 \sum_{k=1}^{n} s_k^3(x_k - x_{k-1}) \\ &\neq c \sum_{k=1}^{n} s_k^3(x_k - x_{k-1}) \\ &= c \int_a^b s(x)\ dx\end{aligned}$$

unless $c \in \{-1, 0, 1\}$

(d) Assume s is defined on $[a, b + c]$. Let $P = \{a = x_0, x_1, \ldots, x_n = b\}$ be a partition of $[a, b]$. Then, observe that $P' = \{a + c = x_0 + c, x_1 + c, \ldots, x_n + c = b + c\}$ is a partition of $[a + c, b + c]$. Now, for $[a + c, b + c]$, we have

$$s(x) = \begin{cases} s_1 & \text{if } x_0 + c < x < x_1 + c \\ s_2 & \text{if } x_1 + c < x < x_2 + c \\ \vdots & \\ s_n & \text{if } x_{n-1} + c < x < x_n + c \end{cases}$$

implying that for $[a, b]$

$$s(x+c) = \begin{cases} s_1 & \text{if } x_0 < x < x_1 \\ s_2 & \text{if } x_1 < x < x_2 \\ \vdots & \\ s_n & \text{if } x_{n-1} < x < x_n \end{cases}$$

Therefore,

$$\begin{aligned} \int_{a+c}^{b+c} s(x)\ dx &= \sum_{k=1}^{n} s_k^3((x_k + c) - (x_{k-1} + c)) \\ &= \sum_{k=1}^{n} s_k^3(x_k - x_{k-1}) \\ &= \int_a^b s(x+c)\ dx \end{aligned}$$

(e) Let $P = \{a = x_0, x_1, \ldots, x_n = b\}$ be a partition of $[a, b]$.

$$\begin{aligned} s(x) < t(x) &\implies s_k < t_k \\ &\implies s_k^3 < t_k^3 \\ &\implies \sum_{k=1}^{n} s_k^3(x_k - x_{k-1}) < \sum_{k=1}^{n} t_k^3(x_k - x_{k-1}) \\ &\implies \int_a^b s(x)\ dx < \int_a^b t(x)\ dx \end{aligned}$$

□

Problem 12

Solution:

(a) Let $P = \{a = x_0, x_1, \ldots, x_j = b, x_{j+1}, \ldots, x_{n-1}, x_n = c\}$ be a partition of $[a, c]$.

$$\begin{aligned} \int_a^b s(x)\ dx + \int_b^c s(x)\ dx &= \sum_{k=1}^{j} s_k(x_k^2 - x_{k-1}^2) + \sum_{k=j+1}^{n} s_k(x_k^2 - x_{k-1}^2) \\ &= \sum_{k=1}^{n} s_k(x_k^2 - x_{k-1}^2) \\ &= \int_a^c s(x)\ dx \end{aligned}$$

(b) Let $P = \{a = x_0, x_1, \ldots, x_n = b\}$ be a partition of $[a, b]$.

$$\begin{aligned}\int_a^b s(x) + t(x)\ dx &= \sum_{k=1}^{n}(s_k + t_k)(x_k^2 - x_{k-1}^2) \\ &= \sum_{k=1}^{n} s_k(x_k^2 - x_{k-1}^2) + \sum_{k=1}^{n} t_k(x_k^2 - x_{k-1}^2) \\ &= \int_a^b s(x)\ dx + \int_a^b t(x)\ dx\end{aligned}$$

(c) Let $P = \{a = x_0, x_1, \ldots, x_n = b\}$ be a partition of $[a, b]$.

$$\begin{aligned}\int_a^b c \cdot s(x)dx &= \sum_{k=1}^{n} cs_k(x_k^2 - x_{k-1}^2) \\ &= c\sum_{k=1}^{n} s_k(x_k^2 - x_{k-1}^2) \\ &= c\int_a^b s(x)\ dx\end{aligned}$$

(d) Assume s is defined on $[a, b + c]$. Let $P = \{a = x_0, x_1, \ldots, x_n = b\}$ be a partition of $[a, b]$. Then, observe that $P' = \{a + c = x_0 + c, x_1 + c, \ldots, x_n + c = b + c\}$ is a partition of $[a + c, b + c]$. Now, for $[a + c, b + c]$, we have

$$s(x) = \begin{cases} s_1 & \text{if } x_0 + c < x < x_1 + c \\ s_2 & \text{if } x_1 + c < x < x_2 + c \\ \vdots & \\ s_n & \text{if } x_{n-1} + c < x < x_n + c \end{cases}$$

implying that for $[a, b]$

$$s(x + c) = \begin{cases} s_1 & \text{if } x_0 < x < x_1 \\ s_2 & \text{if } x_1 < x < x_2 \\ \vdots & \\ s_n & \text{if } x_{n-1} < x < x_n \end{cases}$$

Therefore,

$$\begin{aligned}\int_{a+c}^{b+c} s(x)\ dx &= \sum_{k=1}^{n} s_k((x_k+c)^2-(x_{k-1}+c)^2)\\ &= \sum_{k=1}^{n} s_k(x_k^2-x_{k-1}^2+2c(x_k-x_{k-1}))\\ &\neq \sum_{k=1}^{n} s_k(x_k^2-x_{k-1}^2)\\ &= \int_a^b s(x+c)\ dx\end{aligned}$$

unless $c=0$, $s_k=0$ for all k, or $x_{k-1}=x_k$ for all $1\le k\le n$ (i.e. $a=x_0=x_n=b$).

(e) This property fails. The reason is that the new weight $x_k^2-x_{k-1}^2$ is negative whenever the subinterval lies to the left of 0, so multiplying an inequality by it reverses the inequality.

Take $[a,b]=[-2,-1]$ and let $s(x)=0$ and $t(x)=1$ for all x in $[-2,-1]$. Then $s(x)<t(x)$ for every x in $[a,b]$. Using the partition $P=\{-2,-1\}$,

$$\begin{aligned}\int_{-2}^{-1} s(x)\,dx &= 0\cdot\big((-1)^2-(-2)^2\big)=0,\\ \int_{-2}^{-1} t(x)\,dx &= 1\cdot\big((-1)^2-(-2)^2\big)=1-4=-3.\end{aligned}$$

So

$$\int_{-2}^{-1} s(x)\,dx = 0 > -3 = \int_{-2}^{-1} t(x)\,dx$$

even though $s<t$ on $[-2,-1]$.

Hence the comparison property does not hold in the new theory.

The step $s_k<t_k \implies s_k(x_k^2-x_{k-1}^2)<t_k(x_k^2-x_{k-1}^2)$ is only valid when $x_k^2-x_{k-1}^2>0$, which happens exactly when $0\le x_{k-1}<x_k$. So the property does survive on intervals $[a,b]$ with $a\ge 0$, but not in general

□

Problem 13

Solution:

Let $P = \{a = x_0, x_1, \ldots, x_n = b\}$ be a partition of $[a, b]$ where both s, t are constant on all (x_{k-1}, x_k).

$$\begin{aligned}\int_a^b s(x) + t(x)\ dx &= \sum_{k=1}^{n}(s_k + t_k)(x_k - x_{k-1}) \\ &= \sum_{k=1}^{n} s_k(x_k - x_{k-1}) + \sum_{k=1}^{n} t_k(x_k - x_{k-1}) \\ &= \int_a^b s(x)\ dx + \int_a^b t(x)\ dx\end{aligned}$$

□

Problem 14

Solution:

Let $P = \{a = x_0, x_1, \ldots, x_n = b\}$ be a partition of $[a, b]$ where both s, t are constant on all (x_{k-1}, x_k).

$$\begin{aligned}\int_a^b [c_1 s(x) + c_2 t(x)]\ dx &= \sum_{k=1}^{n}(c_1 s_k + c_2 t_k)(x_k - x_{k-1}) \\ &= \sum_{k=1}^{n} c_1 s_k(x_k - x_{k-1}) + \sum_{k=1}^{n} c_2 t_k(x_k - x_{k-1}) \\ &= c_1 \sum_{k=1}^{n} s_k(x_k - x_{k-1}) + c_2 \sum_{k=1}^{n} t_k(x_k - x_{k-1}) \\ &= c_1 \int_a^b s(x)\ dx + c_2 \int_a^b t(x)\ dx\end{aligned}$$

□

Problem 15

Solution:

Let $P = \{a = x_0, x_1, \ldots, x_n = b\}$ be a partition of $[a, b]$ where both s, t are constant on all (x_{k-1}, x_k).

$$\begin{aligned}s(x) < t(x) &\implies s_k < t_k, \quad \text{for every } (x_{k-1}, x_k) \text{ where } 1 \leq k \leq n \\ &\implies \sum_{k=1}^{n} s_k(x_k - x_{k-1}) < \sum_{k=1}^{n} t_k(x_k - x_{k-1}) \\ &\implies \int_a^b s(x)\ dx < \int_a^b t(x)\ dx\end{aligned}$$

□

Problem 16

Solution:

Let $P = \{a = x_0, x_1, \ldots, x_j = b, x_{j+1}, \ldots, x_{n-1}, x_n = c\}$ be a partition of $[a, c]$.

$$\begin{aligned}\int_a^b s(x)\ dx + \int_b^c s(x)\ dx &= \sum_{k=1}^{j} s_k(x_k - x_{k-1}) + \sum_{k=j+1}^{n} s_k(x_k - x_{k-1}) \\ &= \sum_{k=1}^{n} s_k(x_k - x_{k-1}) \\ &= \int_a^c s(x)\ dx\end{aligned}$$

□

Problem 17

Solution:

Let $P = \{a = x_0, x_1, \ldots, x_n = b\}$ be a partition of $[a, b]$. Then, observe that $P' = \{a + c = x_0 + c, x_1 + c, \ldots, x_n + c = b + c\}$ is a partition of $[a + c, b + c]$. Now, for $[a, b]$, we have

$$s(x) = \begin{cases} s_1 & \text{if } x_0 < x < x_1 \\ s_2 & \text{if } x_1 < x < x_2 \\ \vdots & \\ s_n & \text{if } x_{n-1} < x < x_n \end{cases}$$

implying that for $[a + c, b + c]$

$$s(x - c) = \begin{cases} s_1 & \text{if } x_0 + c < x < x_1 + c \\ s_2 & \text{if } x_1 + c < x < x_2 + c \\ \vdots & \\ s_n & \text{if } x_{n-1} + c < x < x_n + c \end{cases}$$

Therefore,

$$\begin{aligned}\int_a^b s(x)\ dx &= \sum_{k=1}^{n} s_k(x_k - x_{k-1}) \\ &= \sum_{k=1}^{n} s_k((x_k + c) - (x_{k-1} + c)) \\ &= \int_{a+c}^{b+c} s(x - c)\ dx\end{aligned}$$

□